

// GLOBEGRID v5 -- BUILD MANUAL
// DENSE FORMAT -- OPTIMIZED FOR LOW-TOKEN AGENTIC BUILD

GLOBEGRID v5

FIXES, CONTENT QUALITY, AND SCALE

Every item below is a direct fix or build instruction. No filler prose.

REVISION	5.0
FORMAT NOTE	Short prose, bulleted fixes, tables over paragraphs -- written to minimize tokens spent re-deriving intent while building
BUILDS ON	v1-v4, all previously verified present in the live repo

SECTION 01 // CONTENT DISPLAY FIXES

Event title, sorting, history

Fixes

Item	Fix
No raw links in titles	Strip any URL from the generated headline/summary text at synthesis time (v1 s5.5). Raw source links stay in the citation/attribution area only, never inline in the title string.
Time-based sorting	Add explicit sort controls to the live feed and stories directory (v4 s8.2): newest-first (default), oldest-first, most-active-cluster-first.
Past-event access	Add a dedicated 'History' / archive view: paginated, filterable by date range and category, backed by the same permanent fact chain (v1 s5.9) already storing everything -- no new storage needed, just a browsing UI that isn't only 'recent'.

SECTION 02 // LANGUAGE SUPPORT

Expanded language support

Extends v3 s4 (cross-lingual embedding) and v1 s5.8 (display-time translation). Add full UI + translation support for:

- **European:** every EU official language plus Norwegian, Icelandic, Ukrainian, Belarusian, Serbian, Bosnian, Albanian, Macedonian.
- **West/Central Asia and Caucasus:** Persian (Farsi), Turkish, Georgian, Armenian, Azerbaijani, Urdu.
- **ASEAN languages:** Indonesian, Thai, Vietnamese, Filipino/Tagalog, Malay, Burmese, Khmer, Lao.
- **East Asia:** Japanese, Korean, Chinese (Simplified and Traditional as distinct locales, not one 'Chinese' entry).
- **Other:** Swahili, Irish Gaelic, Hebrew, Amharic.

Mechanism unchanged from v1 s5.8: original text/language always preserved; UI chrome and story summaries translate on display. Simplified/Traditional Chinese need separate locale codes since they aren't interchangeable. RTL layout (Persian, Urdu, Hebrew, Arabic already in v2 sources) needs actual RTL CSS support in the panel system (v4 s6), not just translated strings in an LTR layout.

SECTION 03 // CONFLICT VS MILITARY TAGGING

Separate conflict from military-development stories

Currently one bucket. Split into two distinct categories so 'war-scale' stays meaningful.

New: `events.development_type`

Value	Meaning
<code>conflict</code>	Active attacks, territorial changes, ceasefire/negotiation moves -- reserved strictly for literal war-scale developments on an actual conflicts row (v3 s15)
<code>military</code>	Missile tests, drills, war games, force posturing, weapons deals -- tracked and shown, but never auto-tags a <code>conflicts.conflict_id</code> unless it escalates into actual conflict

- Conflict tabs (v3 s15) filter to `development_type = 'conflict'` only -- this is the rule that keeps the conflict tag meaningful.
- Military developments get their own feed filter/category, visible in the main feed and in a country's profile (v3 s13), but not promoted into a conflict tab by themselves.
- Escalation path: if a military-development story later correlates (v1 s5.4) into an active conflict cluster, it's re-tagged conflict -- same auto-suggest/confirm flow already specified in v3 s15.1.

SECTION 04 // TALKDIPLOMACY SOURCE

TalkDiplomacy.com as an ingested source

Add TalkDiplomacy.com to **sources** (v1 s6.1) as a standard RSS/scrape source, same pipeline as any other outlet (v1 s4.1). No special-casing needed -- it's a normal source, just one the project owner controls.

- If TalkDiplomacy has no RSS feed, scrape article listing pages on a schedule instead -- same as any source without native RSS.
- Its own regional article structure (Asia/Europe/etc.) is a natural fit for the country-agenda synthesis (v3 s13.2) as background/historical context, not just current-events input.

SECTION 05 // SOURCE QUALITY

Replacing low-quality GDELT Events volume

GDELT Events (v1 s4.2) is structurally noisy at scale -- high volume, inconsistent quality. Don't rip it out (GDELT Cloud, the structured/clustered layer, stays a primary source per v1 s4.2) -- demote raw GDELT Events to a secondary/backup role and add higher-signal sources on top.

- **Reputable global outlets:** expand v1 s4.1's RSS list with more first-tier wire/broadcast sources beyond the original six.
- **Government/military communications:** add official press feeds per country/department where they exist (defense ministries, foreign ministries, armed forces public affairs) -- same pattern as v3 s18's international-organization sourcing, applied per-country.
- **Reliable social/OSINT feeds:** the LiveUAMap model -- curated, vetted X/social accounts known for fast, accurate ground-level reporting, not a generic keyword-matched social firehose (v2 Reddit integration stays separate and unchanged).
- Weight/rank sources by a simple reliability score (new `sources.reliability_tier`: high/medium/low) so low-tier volume (raw GDELT Events) never drowns out high-tier signal in correlation clustering (v1 s5.4) or the instability index (v1 s5.6).

SECTION 06 // CONFLICT TAB BUG

Fixing empty conflict tabs

Confirmed root cause: `stories.suggested_conflict_id` is read and cleared by the confirm endpoint (`routes_geo.py`) but is never written anywhere in the correlation/extraction pipeline. The auto-suggest half of v3 s15.1 was never actually implemented -- only the confirm/manual half exists. That's why every conflict tab is empty: nothing ever gets suggested in the first place.

- Fix: in the correlation step (v1 s5.4 / `correlate.py`), when a new event/story shares canonical entities (v2 s3.1) with a conflict's registered `conflict_parties` (v3 s15) AND `development_type = 'conflict'` (s3 above), write `suggested_conflict_id` at that point -- this was specified in v3 but the write side was missed.
- Verify end to end after the fix: seed or ingest one real conflict-tagged story, confirm it actually appears in that conflict's tab, not just in the confirm queue.

SECTION 07 // COUNTRY FLAGS

Real flag images, not emoji

Confirmed: no flag rendering exists at all currently (not emoji, not images -- just absent from `CountryPanel.js`). Build this fresh, correctly, the first time.

- Source flag images from Wikimedia Commons (CC-licensed, legal to use with attribution) -- most sovereign-state flags there are SVG, public domain or CC0/CC-BY.
- Add `countries.flag_image_url` (or a locally cached file path after one-time download) -- never render a Unicode flag emoji, which is exactly what was flagged as wrong.
- Cache flags locally after first fetch (small SVG files, ~195 of them) rather than hot-linking Wikimedia on every page load.
- Attribution: flag images get the same credits-page treatment as every other external asset (v4 s22).

SECTION 08 // DEEP ZOOM TERRAIN

Terrain texturing on deep zoom

- As the camera zooms past a threshold, blend in a terrain/elevation texture (heightmap-driven shading, e.g. a public-domain source like NASA's SRTM/Blue Marble data) on the globe surface -- a Paradox-map-style look at close zoom, flat/political coloring at far zoom.
- Toggleable in settings, off by default until proven cheap on the target hardware (ties into v4 s15.4's first-launch performance calibration) -- this is a real rendering cost addition, not free.
- Same LOD philosophy as v4 s2.3's boundary-resolution fix: low detail far away, real detail only where the camera actually is.

SECTION 09 // CLUSTER SPLIT FIX

Fixing 'zoom and click to split' clusters

Confirmed pattern: a 75-event cluster on Kyiv shows a 'zoom and click to split' prompt that doesn't actually work -- the max zoom level available isn't enough to separate 75 points that are geographically this close together.

Zoom-only declustering (v4 s2.2) has a hard ceiling.

- Fix: clicking an unsplittable cluster (screen-space point density still too high at max zoom) opens a scrollable list of its member events directly in the left sliding pane (v4 s17-18) -- same pane system already built for country/story pages, just a list-content template instead of a single-entity template.
- This becomes the actual, reliable way to browse a dense cluster; zoom-based splitting stays as a nice-to-have for less extreme cases, not the only path.

SECTION 10 // RESIZABLE PANELS**Drag-resizable panels**

- Add a drag handle on the inner edge of every docked panel (country/wiki, analyst, event-list per s9 above)
-- standard resize-by-drag, width persisted per-panel-type in local settings so it doesn't reset every session.
- Respect a min/max width so a panel can't be dragged to unusable (too narrow to read) or absurd (covers the whole viewport -- that's what the fullscreen toggle, v4 s6.2, is already for).

SECTION 11 // NON-STATE ACTOR TERRITORY

Rough territory/zone shading

New table: `non_state_actor_zones`

Field	Type	Description
<code>id</code>	<code>uuid PK</code>	
<code>non_state_actor_id</code>	<code>uuid FK</code>	References <code>non_state_actors.id</code> (v3 s16)
<code>zone_geojson</code>	<code>text</code>	Rough polygon(s), deliberately coarse -- not claiming precise operational boundaries
<code>confidence</code>	<code>text</code>	established contested reported -- how solid the claim of presence is
<code>last_updated_at</code>	<code>timestampz</code>	

- Rendered as a shaded, semi-transparent zone on globe/map, toggle-controlled by the existing non-state-actor visibility toggle (v4 s5.4) -- not a separate switch.
- Deliberately rough/approximate polygons, not precise operational detail -- this is descriptive geopolitical context (same standard as ACLED/Liveuamap), consistent with v3 s16's existing scope discipline.

SECTION 12 // 2D MAP PERFORMANCE

Fixing 2D map lag on load

Reported symptom (rendering everything before the user even scrolls) matches a common bug pattern: the map is likely rendering all wrap-around tiles (v4 s4.2) and all data layers at full detail immediately on mount, rather than only what's in or near the current viewport.

- Fix: viewport-culled rendering -- only draw points/borders/labels actually within (or just outside) the current visible bounds, recompute on pan/zoom, not on every frame regardless of view.
- Defer the wraparound tiles (v4 s4.2) -- render the primary tile immediately, generate the east/west repeat tiles lazily just before they'd become visible, not all three eagerly on mount.
- Profile before and after with the browser's own performance tools as part of the fix, not just by feel -- this is exactly the kind of regression that's easy to silently reintroduce later.

SECTION 13 // IMMEDIATE LOAD

Summaries and syntheses load immediately

- Every profile/story page (country, conflict, story) should render its already-computed synthesis text (v3 s13.2, v1 s5.5) instantly from the initial API response -- no separate lazy-loaded follow-up request for content that's already sitting in the database.
- If a synthesis is genuinely still being generated (e.g. brand-new entity, first-ever agenda synthesis job hasn't run), show that state honestly ('synthesis pending') rather than a blank gap that looks broken.
- This is mostly an API-shape check: make sure story/country detail endpoints return everything needed in one response rather than requiring the frontend to make follow-up calls before it can render text that already exists.

SECTION 14 // COLOR THEMES

More toggleable color palettes

- Ship the existing dark-teal theme as default, plus at least: a higher-contrast variant, the colorblind-safe palette already specified in v4 s15.2, and one or two purely aesthetic alternates (e.g. an amber/monochrome 'terminal' look, consistent with the numbers-station audio preset's aesthetic from v4 s12.3).
- Implement as CSS variable swaps, not per-component recoloring -- the whole point of the shared pane/animation system (v4 s18) is that a theme swap should be a small, centralized change.

SECTION 15 // STALE DATA / REFRESH

Fixing outdated leadership data

Confirmed and verified live: `seed_data.py` still lists ("IRN", "head_of_state", "Ali Khamenei (Supreme Leader)", ...). **Checked current facts directly:** Ali Khamenei was assassinated 28 February 2026 during the 2026 Iran war; Mojtaba Khamenei, his son, was confirmed as Iran's new Supreme Leader on 9 March 2026. This one entry is wrong and should be corrected -- but the real fix is the mechanism, not the single row.

- Immediate fix: update the IRN head_of_state row to Mojtaba Khamenei, since = 2026-03-09.
- Real fix: this is exactly what the weekly Wikidata leadership sync (v3 s13.2) is supposed to prevent -- verify that sync job is actually running on schedule, not just specified. A leader who died in February and was replaced in March should never still be showing seed-data content in place of live sync data months later.
- Add a staleness check consistent with v4 s15.3 (visible 'last updated' timestamps): if a leadership record hasn't refreshed within its expected interval, flag it visibly rather than silently serving old data forever.

SECTION 16 // LIGHT BEACON QUALITY

Fixing glitchy event-pin rendering on zoom

- Symptom described ('glitched PS2 texture' look on zoom) is consistent with a low-resolution point-sprite texture being scaled up, or a bloom pass with a mismatched blur radius relative to sprite size at close range.
- Fix: use a resolution-independent approach for the beacon (procedural shader-drawn circle/glow, e.g. signed-distance-field rendering) instead of a fixed-resolution texture, so it stays crisp at any zoom rather than pixelating.
- Decouple glow falloff radius from camera distance properly (recompute in screen-space, not world-space) so the visual size stays consistent and clean whether zoomed in or out -- related to, but distinct from, the click-radius fix in v4 s2.2.

SECTION 17 // AUDIO ENGINE OVERHAUL

Why most presets don't work, and the real fix

Confirmed by inspection: `SoundEngine.js` only has basic oscillators (triangle-wave drone, a generic `osc.type` parameter). There's no distortion/waveshaping, no noise generator, no filter chain. You cannot make metal, industrial, or vaporwave-style sound design with oscillators alone -- this confirms the diagnosis exactly: it's not a tuning problem, it's a missing-primitives problem.

Primitives to add

Item	Fix
<code>WaveShaperNode</code>	Real distortion/saturation -- required for metal, industrial, anything with an edge
<code>Noise generator</code>	Buffered white/pink noise source -- required for static/shortwave (numbers station), percussion transients, texture
<code>BiquadFilterNode chain</code>	Low-pass/high-pass/band-pass shaping -- required for the 'sub bass' and 'filtered pad' character several presets call for
<code>ConvolverNode (reverb)</code>	Space/depth -- currently everything is dry, which reads as flat/cheap regardless of preset
<code>Proper ADSR envelopes</code>	Per-note amplitude shaping -- needed for anything percussive (technohouse, industrial, metal transients) instead of raw on/off oscillator starts

- Rebuild each non-working preset (v4 s12.2-12.3: vaporwave, Dune, metal, technohouse, Berlin industrial, war-room orchestral, data sonification, modal drift) on top of these primitives -- Ambient Default and Arctic/Calm already work because they only ever needed a simple drone, which is the one thing the current engine can actually do.
- Build and verify one preset fully (recommend: Berlin industrial -- simplest to validate the new primitives against) before porting the rest, rather than rebuilding all eight blind.

SECTION 18 // LLM COST

Free and low-cost alternatives to the Claude API

Checked current options directly rather than assuming stale pricing. Real, live options as of mid-2026:

Options

Item	Fix
Anthropic's Claude for Open Source	Launched Feb 2026: qualifying open-source maintainers get 6 months of Claude Max free (~\$1,200 value). GlobeGrid is AGPL-3.0 -- check eligibility first, this may solve the problem directly with the exact model already in use.
Google Gemini API free tier	Gemini 2.5 Flash: 1,500 requests/day, no credit card required. Free-tier prompts are used for training unless billing is enabled -- fine for a personal build, worth knowing before scaling.
Groq	Free tier, ~14,400 requests/day on open-weight models (Llama 3.x), OpenAI-compatible API, very fast. No credit card.
Cerebras	~1M tokens/day free on open-weight models, high throughput, good for batch-style jobs (e.g. the daily briefing, v2 s6.1).
OpenRouter	Aggregator -- one key, routes across many providers' free tiers (including some free Claude/GPT/Gemini slots), useful for stacking multiple free quotas rather than picking just one.
Ollama (local)	Zero cost, no rate limit, fully private -- already the fallback path noted in v1 s3. Trades quality/speed for being genuinely free forever, including after public hosting.

- Recommended path: check Claude for Open Source eligibility first (zero code changes if it works -- same model, same prompts already written). Fall back to Gemini free tier or Groq for anything beyond that quota, using `CLAUDE_MODEL`-style config already in place (v1 s7.2) to swap providers without touching call sites.
- For eventual public hosting: this is the same cost problem noted in v1 Appendix A -- local inference (Ollama) or aggressive free-tier stacking (Gemini + Groq + Cerebras) is the standard path other free tools in this space use, not paying full frontier-API rates per user.

SECTION 19 // SOURCES PAGE BUG

Fixing the un-closeable sources/status page

Likely candidate: the source-status drawer (`StatusPanel.js`) toggles via a single button (`classList.toggle("hidden")`) -- if there's a separate close control (an X button) inside the drawer itself that isn't wired to the same toggle function, that control does nothing, which matches 'can't close once opened.'

- Verify: does the drawer have its own close button distinct from the toggle button that opened it? If so, wire it to call the identical toggle/close function, not a separate no-op.
- Also add: clicking outside the drawer, or pressing Escape, closes it -- standard modal/drawer behavior that should exist regardless of the specific bug found.
- This should get the same shared close/animation behavior as every other panel (v4 s18) rather than a one-off implementation.

SECTION 20 // REGION-AWARE ANALYST

Region-level analyst queries

Extends v3 s24 (analyst panel) and its entity matching (v3 s24.2, corrected in v4 s16): a question like 'what's happening in Eastern Europe?' should resolve to a region, not fail to match or default to one country.

- Add a region-name matcher alongside the existing country/conflict/actor matchers (v3 s24.2 Path 1): match against `countries.region` values (already used for the continent filter, v4 s9.2).
- On a region match: open a new panel content type -- a region summary page listing every country in that region plus an aggregated synthesis of current activity across them, in the same sliding pane (v4 s17-18) as everything else.
- 'Ukraine war' style queries: when the question matches a conflict (v3 s24.2 Path 1, fixed in v4 s16), open that conflict's page and additionally surface its registered `conflict_parties` (v3 s15) as quick links -- Ukraine, Russia, backers, all one click away from the conflict page itself, not a separate lookup.

SECTION 21 // ADDITIONAL IDEAS

A few more, in the same spirit

- **Source reliability display:** surface the new s5 reliability_tier on every citation chip, not just use it internally for ranking -- lets the user see at a glance whether a claim is backed by a high-tier or low-tier source.
- **Panel state persistence:** remember which panels were open, at what size (s10), across app restarts.
- **Diff-on-refresh for synthesized text:** when a country/conflict synthesis regenerates, briefly highlight what changed from the previous version (reuses v3 s12's story version-history diff mechanism, applied to entity syntheses too).
- **Export a conflict's full timeline as a single document:** one-click PDF/markdown export of a conflict's entire tracked history.

SECTION 22 // CONFIG ADDITIONS**New config.yaml values**

```
content:
  strip_links_from_titles: true

sources:
  reliability_tiers_enabled: true
  gdelt_events_role: "secondary"  # was implicitly primary

conflict_tagging:
  development_types: ["conflict", "military"]
  conflict_tab_filter: "conflict"  # military developments never alone populate a tab

globe:
  terrain_texture_enabled: false  # opt-in, calibrate cost first (v4 15.4)
  beacon_render_mode: "sdf"      # was fixed-res sprite

panels:
  resizable: true
  min_width_px: 280
  max_width_px: 900
  persist_across_restarts: true

themes:
  available: ["dark_teal_default", "high_contrast", "colorblind_safe", "amber_terminal"]

llm_provider:
  primary: "anthropic"
  fallback_order: ["anthropic_open_source_grant", "gemini_free", "groq_free", "ollama_local"]

leadership_data:
  staleness_warning_days: 14
```

SECTION 23 // BUILD SEQUENCE

Build order

Phases

Item	Fix
Phase VV	Confirmed one-line/small fixes first: title links (s1), IRN leadership data (s15), sources-page close (s19) -- cheapest, highest-confidence.
Phase WW	Conflict tab bug (s6) + conflict/military split (s3) -- do together, the split's schema is a prerequisite for the auto-suggest fix to tag correctly.
Phase XX	Source quality upgrade (s5) + TalkDiplomacy source (s4) -- both are new source registrations, same pattern, batch together.
Phase YY	Flags (s7) + non-state actor zones (s11) -- both are new visual data layers with a similar asset-sourcing shape (Wikimedia / rough GeoJSON).
Phase ZZ	2D map performance fix (s12) + cluster-split fix (s9) + beacon quality (s16) -- all rendering-correctness fixes, do as one rendering pass.
Phase AAA	Panels: resizable (s10), immediate-load content (s13), region-aware analyst (s20) -- all touch the shared pane system together.
Phase BBB	Audio engine overhaul (s17) -- largest single item here, do in isolation, validate one preset before porting the rest.
Phase CCC	LLM provider fallback (s18) -- independent, mostly config + a provider-swap abstraction.
Phase DDD	Language expansion (s2), deep-zoom terrain (s8), color themes (s14), additional ideas (s21) -- lower urgency, fold in wherever convenient.

// END OF MANUAL -- GLOBEGRID v5 BUILD MANUAL -- REV 5.0